

D903

# Geometry for Secondary Mathematics Teaching

Armored Drill

8 concepts · 12 items · 4 transfer items

**What this is.** Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

**Source.** Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

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## How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

| Target word | The one fact that decides | Answer | [ ] sure [ ] unsure [ ] guess |
|-------------|---------------------------|--------|-------------------------------|
| _____       | _____                     | _____  |                               |

# Items

1.

A teacher sorting quadrilateral cards laminated last year asks which statement is true under inclusive definitions. The cards are stored in a blue bin. Which is true?

- A. Every square is a rhombus.
- B. Every rhombus is a square.
- C. Every trapezoid is a parallelogram.
- D. A square is not a rectangle, because its sides are all equal.

| Target word | The one fact that decides | Answer | [ ] sure [ ] unsure [ ] guess |
|-------------|---------------------------|--------|-------------------------------|
| _____       | _____                     | ___    |                               |

2.

In a dynamic-geometry file shared by a colleague, a triangle is reflected over the line  $x = 1$  and then over the line  $x = 4$ . The file has three saved views. What single transformation is the composition?

- A. A translation 6 units right.
- B. A translation 3 units right.
- C. A translation 6 units left.
- D. A rotation about the point  $(1, 0)$ .

| Target word | The one fact that decides | Answer | [ ] sure [ ] unsure [ ] guess |
|-------------|---------------------------|--------|-------------------------------|
| _____       | _____                     | ___    |                               |

3. [TRANSFER — reversed stem / different variable]

Two reflection lines intersect at a  $35^\circ$  angle. A figure is reflected over the first line, then the second. What is the result?

- A. A reflection over the angle bisector.
- B. A translation.
- C. A rotation of  $70^\circ$  about the intersection point.
- D. A rotation of  $35^\circ$ .

| Target word | The one fact that decides | Answer | [ ] sure [ ] unsure [ ] guess |
|-------------|---------------------------|--------|-------------------------------|
| _____       | _____                     | ___    |                               |

4.

On a town map where 1 unit is 100 meters, a school is at (2, -4) and a library at (8, 6). The map was drawn by the parks department. Where is the point halfway between them?

- A. (5, 5)
- B. (3, 5)
- C. (10, 2)
- D. (5, 1)

| Target word | The one fact that decides | Answer | [ ] sure [ ] unsure [ ] guess |
|-------------|---------------------------|--------|-------------------------------|
| _____       | _____                     | ___    |                               |

5.

Two parallel streets are crossed by a diagonal avenue that was repaved in 2021. One angle formed is  $65^\circ$ . The city has four traffic lights on the avenue. What is the measure of the same-side interior angle paired with it?

- A.  $115^\circ$
- B.  $295^\circ$
- C.  $25^\circ$
- D.  $65^\circ$

| Target word | The one fact that decides | Answer | [ ] sure [ ] unsure [ ] guess |
|-------------|---------------------------|--------|-------------------------------|
| _____       | _____                     | ___    |                               |

6.

A 1:20 scale model of a statue cast in bronze in 1924 needs 0.5 liters of paint to cover its surface. The statue stands in a city plaza. How much paint would the full-size statue need?

- A. 0.025 liters
- B. 10 liters
- C. 200 liters
- D. 4,000 liters

| Target word | The one fact that decides | Answer | [ ] sure [ ] unsure [ ] guess |
|-------------|---------------------------|--------|-------------------------------|
| _____       | _____                     | ___    |                               |

7. [TRANSFER — reversed stem / different variable]

Two similar solids have volumes of  $54 \text{ cm}^3$  and  $16 \text{ cm}^3$ . What is the ratio of their surface areas (larger to smaller)?

- A. 81 : 16
- B. 27 : 8
- C. 9 : 4
- D. 3 : 2

| Target word | The one fact that decides | Answer | [ ] sure [ ] unsure [ ] guess |
|-------------|---------------------------|--------|-------------------------------|
| _____       | _____                     | ___    |                               |

8.

In a circular stained-glass window 2 meters across, an inscribed angle intercepts an arc of  $110^\circ$ . The window was installed by a local artist. What is the inscribed angle's measure?

- A.  $220^\circ$
- B.  $110^\circ$
- C.  $70^\circ$
- D.  $55^\circ$

| Target word | The one fact that decides | Answer | [ ] sure [ ] unsure [ ] guess |
|-------------|---------------------------|--------|-------------------------------|
| _____       | _____                     | ___    |                               |

9. [TRANSFER — reversed stem / different variable]

An inscribed angle measures  $38^\circ$ . What is the measure of its intercepted arc?

- A.  $38^\circ$
- B.  $76^\circ$
- C.  $142^\circ$
- D.  $19^\circ$

| Target word | The one fact that decides | Answer | [ ] sure [ ] unsure [ ] guess |
|-------------|---------------------------|--------|-------------------------------|
| _____       | _____                     | ___    |                               |

10.

On a school-issued laptop with a sticker-covered lid, a student builds a “square” in a dynamic-geometry app by placing four segments by eye. It looks perfect on screen. How should the teacher check whether it is a true construction?

- A. Accept it, since it looks perfect.
- B. Drag a vertex and see whether the figure stays a square.
- C. Print it and check the sides with a ruler.
- D. Measure the sides once in the current position.

| Target word | The one fact that decides | Answer | [ ] sure [ ] unsure [ ] guess |
|-------------|---------------------------|--------|-------------------------------|
| _____       | _____                     | ___    |                               |

11.

A stop sign on the corner outside the school is a regular octagon. The sign is 75 cm across. What is the measure of each interior angle?

- A.  $157.5^\circ$
- B.  $45^\circ$
- C.  $1,080^\circ$
- D.  $135^\circ$

| Target word | The one fact that decides | Answer | [ ] sure [ ] unsure [ ] guess |
|-------------|---------------------------|--------|-------------------------------|
| _____       | _____                     | ___    |                               |

**12.** [TRANSFER — reversed stem / different variable]

A polygon's interior angles sum to  $1,440^\circ$ . How many sides does it have?

- A.** 6
- B.** 10
- C.** 8
- D.** 4

| Target word | The one fact that decides | Answer | [ ] sure [ ] unsure [ ] guess |
|-------------|---------------------------|--------|-------------------------------|
| _____       | _____                     | _____  |                               |

# STOP

Answers and dissection follow. Finish every item and every strip first.

# Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

## 1. Answer A Concept 1

A teacher sorting quadrilateral cards ~~laminated last year~~ asks which statement is true under inclusive definitions. ~~The cards are stored in a blue bin.~~ Which is true?

**Target word:** inclusive classification **Decides:** A square meets the rhombus definition: four congruent sides.

|   |                    |                                                                                  |
|---|--------------------|----------------------------------------------------------------------------------|
| A | <b>Correct</b>     | A square has four congruent sides, which is the definition of a rhombus.         |
| B | Inverted operation | Reverses the inclusion; a rhombus need not have right angles.                    |
| C | Adjacent category  | A trapezoid may have only one pair of parallel sides.                            |
| D | Confusion set      | Uses exclusive categories; a square has four right angles, so it is a rectangle. |

## 2. Answer A Concept 2

In a dynamic-geometry file ~~shared by a colleague~~, a triangle is reflected over the line  $x = 1$  and then over the line  $x = 4$ . ~~The file has three saved views.~~ What single transformation is the composition?

**Target word:** composition of reflections **Decides:** Parallel lines 3 units apart.

|   |                    |                                                                           |
|---|--------------------|---------------------------------------------------------------------------|
| A | <b>Correct</b>     | Twice the 3-unit gap, in the direction from the first line to the second. |
| B | Dropped element    | Leaves out the factor of two.                                             |
| C | Inverted operation | Moving from $x = 1$ to $x = 4$ means the shift is to the right.           |
| D | Confusion set      | Rotations come from intersecting lines, not parallel ones.                |

## 3. Answer C Concept 2 · transfer

Two reflection lines intersect at a  $35^\circ$  angle. A figure is reflected over the first line, then the second. What is the result?

**Target word:** composition over intersecting lines **Decides:** Rotation of twice the angle.

|   |                   |                                                                                  |
|---|-------------------|----------------------------------------------------------------------------------|
| A | Adjacent category | Two reflections preserve orientation, so the result cannot be a reflection.      |
| B | Confusion set     | Translations come from parallel lines.                                           |
| C | <b>Correct</b>    | Two reflections over intersecting lines make a rotation of $2 \times 35^\circ$ . |
| D | Dropped element   | Leaves out the factor of two.                                                    |

## 4. Answer D Concept 3

On a town map ~~where 1 unit is 100 meters~~, a school is at  $(2, -4)$  and a library at  $(8, 6)$ . ~~The map was drawn by the parks department.~~ Where is the point halfway between them?

**Target word:** midpoint **Decides:** Average the coordinates.

|   |                    |                                                              |
|---|--------------------|--------------------------------------------------------------|
| A | Confusion set      | Drops the negative sign on $-4$ .                            |
| B | Inverted operation | Halves the differences instead of averaging the coordinates. |
| C | Dropped element    | Adds but forgets to divide by 2.                             |
| D | <b>Correct</b>     | $(2 + 8)/2 = 5$ ; $(-4 + 6)/2 = 1$ .                         |

## 5. Answer A Concept 4

Two parallel streets are crossed by a diagonal avenue ~~that was repaved in 2021~~. One angle formed is  $65^\circ$ . ~~The city has four traffic lights on the avenue.~~ What is the measure of the same-side interior angle paired with it?

**Target word:** same-side interior angle **Decides:** Same-side interior angles are supplementary.

|   |                    |                                                               |
|---|--------------------|---------------------------------------------------------------|
| A | <b>Correct</b>     | $180^\circ - 65^\circ = 115^\circ$ .                          |
| B | Inverted operation | Subtracts from $360^\circ$ instead of $180^\circ$ .           |
| C | Adjacent category  | Uses the complement ( $90^\circ$ ) instead of the supplement. |
| D | Confusion set      | Treats same-side interior angles as congruent.                |

## 6. Answer C Concept 5

A 1:20 scale model of a statue ~~cast in bronze in 1924~~ needs 0.5 liters of paint to cover its surface. ~~The statue stands in a city plaza.~~ How much paint would the full-size statue need?

**Target word:** area scaling **Decides:** Paint covers area, so multiply by  $20^2 = 400$ .

|   |                    |                                        |
|---|--------------------|----------------------------------------|
| A | Inverted operation | Divides by 20 instead of multiplying.  |
| B | Dropped element    | Scales by $k$ , not $k^2$ .            |
| C | <b>Correct</b>     | $0.5 \times 400 = 200$ .               |
| D | Confusion set      | Scales by $k^3$ , which is for volume. |

## 7. Answer C Concept 5 · transfer

Two similar solids have volumes of  $54 \text{ cm}^3$  and  $16 \text{ cm}^3$ . What is the ratio of their surface areas (larger to smaller)?

**Target word:** area ratio from volume ratio **Decides:** Volume ratio  $27/8$  means  $k = 3/2$ .

|   |                    |                                                           |
|---|--------------------|-----------------------------------------------------------|
| A | Inverted operation | Raises $k$ to the 4th power.                              |
| B | Dropped element    | That is the volume ratio.                                 |
| C | <b>Correct</b>     | $k = \sqrt[3]{(27/8)} = 3/2$ ; area ratio $= k^2 = 9/4$ . |
| D | Adjacent category  | That is the length ratio.                                 |

## 8. Answer D Concept 6

In a circular stained-glass window ~~2 meters across~~, an inscribed angle intercepts an arc of  $110^\circ$ . ~~The window was installed by a local artist.~~ What is the inscribed angle's measure?

**Target word:** inscribed angle **Decides:** Half the intercepted arc.

|   |                    |                                                                     |
|---|--------------------|---------------------------------------------------------------------|
| A | Inverted operation | Doubles instead of halves.                                          |
| B | Confusion set      | That is the central angle on the same arc.                          |
| C | Adjacent category  | $180^\circ - 110^\circ$ uses a supplement rule that does not apply. |
| D | <b>Correct</b>     | $110^\circ \div 2 = 55^\circ$ .                                     |

### 9. Answer B Concept 6 · transfer

An inscribed angle measures  $38^\circ$ . What is the measure of its intercepted arc?

**Target word:** solve for the arc **Decides:** Arc =  $2 \times$  inscribed angle.

|          |                    |                                                    |
|----------|--------------------|----------------------------------------------------|
| A        | Dropped element    | Leaves out the factor of two.                      |
| <b>B</b> | <b>Correct</b>     | $2 \times 38^\circ = 76^\circ$ .                   |
| C        | Adjacent category  | $180^\circ - 38^\circ$ is not an arc relationship. |
| D        | Inverted operation | Halves again instead of doubling.                  |

### 10. Answer B Concept 7

On a school-issued laptop ~~with a sticker covered lid~~, a student builds a “square” in a dynamic-geometry app by placing four segments by eye. It looks perfect on screen. How should the teacher check whether it is a true construction?

**Target word:** construction vs. drawing **Decides:** A construction keeps its properties when dragged.

|          |                   |                                                              |
|----------|-------------------|--------------------------------------------------------------|
| A        | Camouflage        | How it looks in one position is not evidence.                |
| <b>B</b> | <b>Correct</b>    | A drawing falls apart when dragged; a construction does not. |
| C        | Adjacent category | Still a single static check.                                 |
| D        | Dropped element   | A careful drawing can pass one static check.                 |

### 11. Answer D Concept 8

A stop sign ~~on the corner outside the school~~ is a regular octagon. ~~The sign is 75 cm across.~~ What is the measure of each interior angle?

**Target word:** interior angle of a regular polygon **Decides:**  $(8 - 2) \times 180^\circ = 1,080^\circ$ .

|          |                   |                                                                      |
|----------|-------------------|----------------------------------------------------------------------|
| A        | Confusion set     | $(8 - 1) \times 180^\circ \div 8$ uses the wrong count of triangles. |
| B        | Adjacent category | That is the exterior angle ( $360^\circ \div 8$ ).                   |
| C        | Dropped element   | The total, not divided among the 8 angles.                           |
| <b>D</b> | <b>Correct</b>    | $1,080^\circ \div 8 = 135^\circ$ .                                   |

### 12. Answer B Concept 8 · transfer

A polygon’s interior angles sum to  $1,440^\circ$ . How many sides does it have?

**Target word:** solve for n **Decides:**  $(n - 2) \times 180 = 1,440$ .

|          |                    |                                                  |
|----------|--------------------|--------------------------------------------------|
| A        | Inverted operation | Subtracts 2 instead of adding.                   |
| <b>B</b> | <b>Correct</b>     | $1,440 \div 180 = 8 = n - 2$ , so $n = 10$ .     |
| C        | Dropped element    | Forgets to add the 2 back.                       |
| D        | Confusion set      | Divides by $360^\circ$ , the exterior-angle sum. |

## Triage grid

Write each item number in the box it belongs to.

|                       | <b>Right</b>                                          | <b>Wrong</b>                                                                          |
|-----------------------|-------------------------------------------------------|---------------------------------------------------------------------------------------|
| <b>Sure</b>           | Owled.                                                | <b>Concept inversion — top priority.</b> A wrong mental model; it survives rewording. |
| <b>Unsure / guess</b> | Recognition, not mastery. Re-drill with the transfer. | Not learned yet.                                                                      |

## Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

| <b>Trap</b>        | <b>What it does</b>                                          | <b>In this drill</b> | <b>Your misses</b> |
|--------------------|--------------------------------------------------------------|----------------------|--------------------|
| Dropped element    | Leaves out a required term or step.                          | 9                    |                    |
| Wrong stem number  | A figure copied from the prompt that is not the answer.      | 0                    |                    |
| Adjacent category  | The right idea in the wrong bucket.                          | 8                    |                    |
| Inverted operation | Direction flipped: add/subtract, cause/effect, higher/lower. | 9                    |                    |
| Absolute language  | Always, never, guarantee — where judgment applies.           | 0                    |                    |
| Confusion set      | Every option is real; the test is exact role-matching.       | 9                    |                    |
| Camouflage         | Answers a stem fact that was never the question.             | 1                    |                    |

# Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

| # | Concept                                                  | Rule                                                                                                                                                                      | Trap                                                              | Tell                                                            |
|---|----------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------|-----------------------------------------------------------------|
| 1 | <b>Inclusive classification of quadrilaterals</b>        | Definitions are inclusive: a square is a rectangle and a rhombus, and a rectangle is a parallelogram, because each meets the more general definition.                     | Treating categories as exclusive (“a square is not a rectangle”). | “Every ___ is a ___” statements about quadrilaterals.           |
| 2 | <b>Composing reflections</b>                             | Reflecting across two parallel lines is a translation of twice the distance between them; across two intersecting lines it is a rotation of twice the angle between them. | Forgetting the factor of two, or expecting another reflection.    | “Reflect over one line, then another.”                          |
| 3 | <b>Midpoint formula</b>                                  | The midpoint is the average of the endpoints: $((x_1 + x_2)/2, (y_1 + y_2)/2)$ .                                                                                          | Subtracting the coordinates, or forgetting to halve.              | “Halfway between” two coordinate points.                        |
| 4 | <b>Angles formed by parallel lines and a transversal</b> | Corresponding and alternate interior angles are congruent; same-side interior angles are supplementary.                                                                   | Calling all related angles congruent.                             | Parallel lines, a transversal, and “same-side interior.”        |
| 5 | <b>Scaling area and volume</b>                           | If lengths scale by $k$ , areas scale by $k^2$ and volumes by $k^3$ .                                                                                                     | Scaling area or paint by $k$ , or area by $k^3$ .                 | A scale model and a question about surface, paint, or capacity. |
| 6 | <b>Inscribed angle theorem</b>                           | An inscribed angle is half its intercepted arc (and half the central angle on that arc).                                                                                  | Setting the inscribed angle equal to the arc.                     | An angle with its vertex on the circle.                         |
| 7 | <b>Constructions survive the drag test</b>               | In dynamic geometry, a true construction keeps its defining properties when points are dragged; a drawing only looks right in one position.                               | Accepting a figure because it looks right or measures right once. | A figure built in dynamic software and “how do you check it?”   |
| 8 | <b>Polygon interior angle sum</b>                        | The interior angles of an $n$ -gon sum to $(n - 2) \times 180^\circ$ ; each angle of a regular $n$ -gon is that sum divided by $n$ .                                      | Using $n$ instead of $n - 2$ , or reporting the exterior angle.   | A polygon and its number of sides or angle sum.                 |

**Course scope (public description):** Geometry for Secondary Mathematics Teaching is a comprehensive course designed to equip future educators with the expertise to teach secondary-level geometry. This course delves into core geometric concepts, including geometric reasoning, transformations, congruence, similarity, and coordinate geometry, while emphasizing the development of critical thinking and problem-solving skills. Students will engage in dynamic instructional strategies and hands-on learning experiences, using tools like GeoGebra and Desmos to create interactive lessons that address common misconceptions and support diverse learners. By the end of the course, students will have the confidence and skills to design, implement, and refine geometry lessons aligned with educational standards and tailored to the needs of a diverse student population, preparing them to inspire and engage the next generation of mathematicians.